	Academic Semester	Fall 2026-2027	**Academic level**	1
	Course Code	MA104	**Exam**	Quiz 1 — Model G
	Course Title	Discrete Mathematics		
	Time Allowed	30 Minutes	**Total Points**	10
	Course Instructors	Dr. Salah Moawad , Dr. Tamer Seuodi		

Student Name: _____ **Student ID:** _____ **Section:** _____

*Answer **all** questions. Use T / F in truth tables and 1 / 0 in membership tables.*

Question 1 — Choose the correct answer and give the reason (5 points)

One point each: circle the letter of the correct answer **and** write a short reason in the box.

1. If p is **F** and q is **F**, which proposition is **false**?

A) $p \to q$ **B)** $p \leftrightarrow q$
C) $p \vee q$ **D)** $\neg p \wedge \neg q$

Reason:

2. Let $A = \{1, 5, 7\}$ and $B = \{5, 7, 9\}$. Then $B - A =$

A) $\{9\}$ **B)** $\{1\}$
C) $\{5, 7\}$ **D)** $\{1, 9\}$

Reason:

3. $\neg(p \to q) \equiv$?

A) $\neg p \to \neg q$ **B)** $p \wedge \neg q$
C) $\neg p \vee q$ **D)** $\neg p \wedge q$

Reason:

4. Is $((\neg p) \vee (q \to r))$ a **well-formed formula** (WFF)? If not, name the rule it breaks.

A) WFF **B)** Not a WFF

Reason:

5. Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{2, 3, 4\}$, $B = \{4, 5\}$. Then $A^c \cup B =$

A) $\{4\}$ **B)** $\{1, 4, 5, 6\}$
C) $\{1, 5, 6\}$ **D)** $\{2, 3\}$

Reason:

Question 2 — Truth table (2.5 points)

(a) Complete the truth table of $(p \wedge \neg q) \vee (\neg p \wedge q)$. [2 pts]

p	q	$\neg p$	$\neg q$	$p \wedge \neg q$	$\neg p \wedge q$	$(p \wedge \neg q) \vee (\neg p \wedge q)$
T	T					
T	F					
F	T					
F	F					

(b) Let P be the proposition in (a). Is $P \equiv p \oplus q$? Justify using your table. [½ pt]

Question 3 — Membership table (2.5 points)

(a) Complete the membership table for the identity $A \cup (B - A) = A$. [2 pts]

A	B	$B - A$	$A \cup (B - A)$	A
1	1			
1	0			
0	1			
0	0			

(b) Is the identity $A \cup (B - A) = A$ **true** or **false**? Justify using your table. [½ pt]

— End of quiz · Good luck —